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# Implementation strategies supporting fall prevention interventions in a long-term care setting for older persons: a systematic review protocol



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# Implementation strategies supporting fall prevention interventions in a long-term care setting for older persons: a systematic review protocol

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## Abstract

**Introduction:** Falls are common among older people in long-term care settings (LCS). Falls lead to significant morbidity, mortality, and reduced quality of life among residents. Fall prevention interventions have been shown to reduce falls in LCS. However, this may not always translate to effectiveness in real-world situations. We will conduct a systematic review (SR) to identify the implementation strategies used in fall prevention interventions in LCS, examining the effectiveness of strategies in terms of key implementation outcomes and falls reduction.

**Methods and analysis**: The search will include scientific papers in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, and published theses. The SR will consider all original research that has empirically evaluated or tested implementation strategies to support fall prevention interventions in LCS, published in English or Arabic between 1st January 2001 and 1st May 2021, where data is presented on the implementation strategy or outcome. Clinical trials, quasi-experimental studies and quality improvement studies will be eligible for



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inclusion. Two independent researchers will complete screening, data abstraction and quality assessments. The screening process will be presented using a PRISMA flow diagram. Data will be extracted into a standardised table, including the country, year, authors, type of study,clinical outcome (falls rate and/or risk reduction as available), implementation strategy, and implementation outcomes. Implementation strategies will be defined and categorised using the Expert Recommendation for Implementing Change Taxonomy. Implementation outcomes will be defined and categorised using the Implementation Outcomes Taxonomy, and clinical outcomes of the intervention's effectiveness for fall preventions will be reported as formulated in each study, with a final narrative synthesis of the data.

**Ethics and dissemination:** The results of the systematic review will be disseminated via peer-reviewed journals, published in open-access journals and presented at international conferences.

**Trial registration:** This SR was registered on the PROSPERO database; registration number: CRD42021239604 (30 April 2021).

**Keywords**: Falls prevention, older person, long-term care, residential care, implementation, feasibility.

**Strength and limitation of this study**:

- This planned review of effective implementation strategies and processes used in falls interventions in LCS will help to bridge the evidence-practice gap and to facilitate successful implementation of these interventions in the real world.
- This planned review will Identify the range of implementation strategies used in fall prevention interventions in LCS, and the process of implementing the intervention.
- Data concerning implementation strategies and outcomes will be categorised and defined using the Expert Recommendation for Implementing Change Taxonomy and the implementation outcomes taxonomy frameworks, which are considered seminal in the field of implementation science.
- The planned review will include all the clinical trials types, and it is not applicable to be conducted as a meta-analysis.
- The results of this planned review will be summarised to provide a descriptive review,



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**Background and rationale**

Falls are considered a serious health issue, with the World Health Organisation (WHO) reporting that more than 30% of people aged 65 or older experience at least one fall yearly, while 50% of these experience recurrent falls (1). Falls can lead to decreased physical activity and loss of independence, increased fear of falling and depression, and reduced confidence and life quality among older people (2,3). The incidence rate of falls in long-term care settings (LCS) is reported to be 1.7 falls per year per resident, more than three times that of community dwellers (4). Half of all LCS residents experience falls each year, while 25% to 30% suffer physical injuries after falling (5,6). Hip fracture due to falls is estimated to occur in 3% to 5% of residents annually in LCS (7). Fall-related injuries substantially increase morbidity, mortality and hospitalisation rates among residents in LCS (6,8). Indeed, 40% of mortality in residential care is caused by fall injuries (9).

Preventing falls among LCS residents is considered a challenge, and many fall prevention interventions have been described, whether single, multicomponent (standardised) or multifactorial (tailored) interventions (10,11). These latter fall prevention interventions typically include exercise, environmental assessment, staff education and training, medication review, vitamin D supplementation, etc. (2). Several systematic reviews (SR) and meta-analyses (MA) have evaluated the effects and efficacy of fall prevention interventions in LCS (2,3,11–13). An SR and MA of 14 studies conducted up to 2013 reported that the rate of recurrent falls decreased by 21% due to fall prevention interventions, although there was no reduction in the overall number of falls (2). It also reported that multifactorial interventions significantly reduced the number of falls and recurrent falls, whilst single and multicomponent interventions did not. This same SR reported that staff education and training as a single intervention had a possibly harmful effect by increasing the fall rate in the intervention groups compared to the control groups (2). A later SR of 36 studies conducted up to 2019 reported that fall prevention interventions have a beneficial effect in reducing falls, the number of residents with one fall during the intervention and follow-up, and the number of



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recurrent fallers by 27%, 20% and 30%, respectively, in LCS residents (11). It also reported that all multifactorial interventions effectively reduced the number of fallers, while single interventions substantially decreased the number of recurrent fallers. As a single component, exercise is effective, reducing by 36% the number of fallers and by 41% the number of recurrent fallers. Some single interventions involving staff education and training demonstrated benefits in terms of reducing falls and recurrent falls, whereas others were not effective (11).

Existing evidence indicates to a degree how multifactorial interventions effectively prevent falls in LCS (14,15), in that they are comprehensively tailored to residents' needs by including multidimensional intervention components and are provided by a multidisciplinary team focusing on assessing and treating the many factors that lead to falls (8). However, implementing these interventions and other kinds of fall prevention interventions in a real-world situation may be complex and challenging, because they contain many components and require change at multiple levels. In an SR of mixed-method studies which aimed to identify factors that limited fall programme success in LCS, there were 27 barriers, with the following being the most common: staff feeling overwhelmed, helpless, staffing issues, limited knowledge and skills, frustration and concern about the ability to control fall management (15). The SR authors recommended staff training to improve communication, knowledge and skills, which are modifiable factors when developing a fall prevention programme (15), and they suggested that multifactorial intervention should address barriers to implementation.

The intervention's effectiveness might be influenced by the implementation strategy used and how well the intervention is implemented, which is defined as "the methods and techniques used to enhance the adoption, implementation, and sustainability of a clinical programme or practice" (16). The broader-implementation scientific literature highlights the effectiveness of some implementation strategies (e.g., audits and feedback, educational outreach, practice facilitation, local opinion leaders, etc.) (17). The implementation outcomes are considered precursors to service



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and health outcomes, and they are defined as "the effects of deliberate and purposive actions to implement new treatments, practices, and services" (18), as they achieved give a good indication of the extent and quality of the implementation of the intervention. However, all previous SRs reported only sparsely on the implementation strategies or outcomes of the fall prevention interventions. Therefore, determining the process and success of fall prevention intervention implementation is a key issue to be addressed.

We aim to address the current evidence-practice gap by identifying the range of implementation strategies used in fall prevention interventions for older people in LCS, indicating their level of success as measured by implementation outcomes and fall reduction.

**Research question**

- What implementation strategies have been used in single component or multicomponent/multifactorial fall prevention interventions in LCS to date?
- What implementation strategies are most successful in terms of implementation outcomes and falls prevention?

**Objective**

This SR aims to synthesise the evidence on implementation strategies used to support falls prevention interventions (including RCTs of all types, quality improvement projects and quasi-experimental studies) as follows:

- Identify the range of implementation strategies used in fall prevention interventions in LCS, and the process of implementing the intervention.
- Assess the effectiveness of these strategies used to implement fall prevention interventions in LCS.

**Methods**



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This review was registered on the International Prospective Register of Systematic Reviews (PROSPERO) database (registration number CRD42021239604). The reporting of the review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol (PRISMA-P). The PICO framework was used for framing and reporting the inclusion and exclusion criteria for this protocol.

**Types of participants/population**

This review will consider all the studies that had an intervention directed at all staff members who worked at LCS for older people aged 65 years or older and residing in LCS. LCS is defined as "residential facilities that provide 24-hour-a-day surveillance, personal care, and limited clinical care for persons who are typically elderly and infirm" (11). This includes nursing homes and care homes. We will exclude studies reporting on specific populations in LCS (e.g., those residing in long-stay mental health facilities) and studies involving only people with cognitive issues, intellectual disability, etc. We will include studies aimed at mixed populations in LCS, where the data related to older people can be separated.

**Types of interventions**

This review will consider all studies aimed at preventing falls, whether designed as single, multi-component or multifactorial interventions, and whether delivered by multidisciplinary teams or by a single discipline. We will include all studies that describe the use of any implementation strategy, such as champions/local leaders, audits/feedback, education materials, workshops, etc. Strategies will be defined using the Expert Recommendation for Implementing Change Taxonomy (ERIC) (19,20)

**Comparisons**

This will include any other interventions or usual care.

**Types of outcomes**



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This review will consider all the categories of outcomes (i.e., implementation outcomes, service outcomes and client outcomes (18)). There will not be restrictions on studies based on the outcomes, and we will examine the kinds of outcomes that have been measured. Our primary interest is the implementation outcome, including such aspects as the feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration of the intervention, as per Proctor et al. (18). We will also report on the effect of the intervention on clinical outcomes (e.g., as regards falls reduction), noting that this is reported in intervention studies as falls risk reduction, falls rate reduction, time to first fall, etc.

**Types of studies**

We will consider all intervention studies involving experimental and quasi-experimental designs, including randomised controlled trials (single site, cluster), feasibility studies for randomised controlled trials, quality improvement empirical studies, and pre-post test empirical designs. In addition, our primary focus is the implementation strategy, and we anticipate this to involve quantitative, qualitative and mixed data; we will include the feasibility studies combined mixed data. All relevant studies published in English or Arabic in the last 20 years (beginning 1st January 2001) will be included, noting that implementation science is a relatively new field so that we expect little data to be available prior to 2000.

*Table 1: Inclusion and exclusion criteria according to the PICO framework*

| **PICO framework** | **Inclusion criteria** | **Exclusion criteria** |
|---|---|---|
| **Population** | • All staff members in LCS who have worked with older people aged 65 or above.<br>• Studies involving mixed populations where the older population is studied separately, based in a long-care setting or mixed settings (if LCS-related | • Not directed at the staff members in LCS<br>• Studies intervened for only individuals aged under 65, or where data is not separately reported for the older people within a mixed-age population<br>• Study based outside of an LCS<br>• Studies relating to specific populations in LCS, e.g. long-stay mental health |



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|---|---|---|
| | data can be separated from other data). | facilities, studies only of people with cognitive issues, intellectual disability, etc<br>• Non-English, non-Arabic language<br>• Published before 2001 |
| **Interventions** | • Fall prevention interventions, whether a single component or multifactorial/multicomponent, where there is an implementation strategy or process described | • Studies where the implementation strategy or process is not described |
| **Comparator** | Usual care or other interventions | There will be no restriction on the comparator used in eligible studies |
| **Outcomes** | All categories of outcomes (i.e., implementation outcomes, service outcomes, client outcomes) | There will be no restriction on the outcomes. |

**Information resource and search strategy**

We will conduct searches for all published interventions studies (as specified above) in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, limited to the last 20 years. The search will also include published theses within the last 20 years, identified using the following grey literature databases:

- OPEN GREY (www.opengrey.eu)
- Open Access Theses and Dissertations (OATD) (www. Oatd.org)
- ProQuest (www. Proquest.com)
- British library EThOS (www.ethos.bl.uk).
- EBSCO Open dissertation (www.ebsco.com)
- RIAN (www.rian.ie)
- LENUS (www. lenushealth.com)
- CORA (www. cora.ucc.ie)

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The process of searching will combine free and controlled terms, including text-specific keywords or medical subject headings (MeSH) terms, and it will be integrated with Boolean operators. The search will be filtered to include studies in the English and Arabic languages, in an LCS, all intervention study types, and older people aged 65 years and above (Table 1). A forward and backward citation search of the included studies will be performed. We will also conduct a manual search, using the names of the authors of intervention studies, seeking any linked feasibility or implementation papers.

*Table 2: Search term strategy used in electronic databases*

| | |
|---|---|
| **1- population and settings** | "long-term care setting*" OR "nursing home*" OR "residential care setting*" OR "Residential facilit*" OR "institution care" OR "nursing care" OR "nursing facilit*" OR "continuous care" |
| **2- Intervention/ phenomena of interest** | Faller* OR "accidental fall*" OR falling OR falls OR slip* OR "fall prevention" OR "falls prevention"<br>**AND**<br>Prevent* OR reduc* OR minimi* OR decreas* OR interven* |
| **3- outcome** | Feasib* OR sustain* OR adopt* OR accept* OR appropriate* OR fidelity OR implement* OR uptake* OR adher* OR facilitat*OR barrier* OR accessib* OR penetrat* OR mechanism* OR mediat* OR driv* |
| **Combination with Boolean operators** | 1 AND 2 AND 3 |
| **limitation** | English and Arabic languages<br>Published since 2001 |

**Data management and selection process**

All retrieved articles will be imported and managed using the evidence synthesis software Covidence (www.covidence.org), with full references included. Duplicate studies will be removed using the application. Two independent reviewers will screen the retrieved studies based on titles and abstracts, discussing any discrepancies and agreeing with a list of potential studies for full screening. The two reviewers will then examine the full text of these articles independently to ensure that they fulfil the inclusion criteria. Disagreement about eligibility at this stage will be resolved through



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discussion and consultation with an additional reviewer if necessary. The results of the search strategy and selection of studies for inclusion will be reported using a PRISMA flow diagram.

**Data Extraction**

Two independent reviewers will extract the data regarding the eligible studies. The data will then be compared, and any disagreement will be solved by discussing until a consensus is reached, with input from a third (senior) researcher if needed.

The data extracted will include the following details:

- First Author, Publication year, Country
- Study design
- Participant eligibility criteria and sample size
- Fall intervention characteristics
- Implementation theory or framework
- Implementation strategy (type, number, target, temporality, dose)
- Implementation outcomes (e.g. fidelity)
- Clinical outcome (fall reduction as described in the study- e.g. relative risk, relative rate, time to first fall, etc.)

**Data synthesis procedure**

Data concerning implementation strategies and outcomes will be categorised and defined using Powell et al.'s and Proctors et al.'s frameworks, which are considered seminal in the field of the implementation science (18,21). Powell et al. distinguished implementation strategies using the Consolidated Framework for Implementation Research. A group of researchers and expert clinicians then worked to develop and revise Powell's list of implementation strategies (19). These implementation strategies were updated, reaching 73 in number, and titled the Expert Recommendation for Implementing Change Taxonomy (ERIC), and these strategies are grouped into



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nine categories (19,20). The implementation strategy data for this review will be characterised under 73 distinct strategies and then be synthesised under these nine subheadings.

Proctor et al. developed an implementation outcomes taxonomy (18). This taxonomy distinguishes the concept of implementation outcomes from service system outcomes and clinical treatment outcomes. It provides definitions of eight implementation outcomes: feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration. The implementation outcome data of this review will be coded using the Proctor taxonomy and synthesised.

**Quality appraisal**

Two independent reviewers will assess the quality of the studies included. Any disagreement will be discussed and resolved via a third reviewer. We expect that some intervention studies will embed implementation data within the main clinical outcome paper, and others will report this as a separate paper.

All RCTs and quasi-experimental studies will be assessed using the relevant checklists from the Joanna Briggs Institute (JBI ) Critical Appraisal Tool for determining the quality of the studies (22,23). Each criterion is rated as “Yes” if the criteria are clearly reported, “NO” if not, “Unclear” if there is no information, or “Not Applicable” if not relevant to the study (24).

The Mixed Method Appraisal Tool (MMAT) will be used to assess the quality of mixed data (25). MMAT consists of five categories based on the study designs. It is rated with a “Yes” if the criteria are clearly met, or a “NO” if not; “can't tell” can be selected if there is unclear information on the reporting relating to the criterion. The authors have suggested calculating an overall score by scoring the presence/absence of each criteria as 1/0 and then dividing the sum of “presence” responses by the number of “relevant criteria” and multiplying this by 100 (26).

**Patient and public involvement:**



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No patient and public will be involved in this review.

**Summary of evidence and reporting of the review**

The results will be summarised to provide a descriptive review of effective implementation strategies for interventions aimed at preventing falls in a long-term care setting, both in terms of leading to successful clinical outcome and in terms of implementation success. The reporting of the review will follow Systematic Reviews and MetaAnalyses (PRISMA) guidelines.

**Abbreviations**

LCS: long-term care setting; WHO: The World Health Organisation; SR: systematic review; RCT: randomised controlled trial; PROSPERO: The International Prospective Register of Systematic Reviews; PRISMA-P: The Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol; MeSH: Medical subject headings; ERIC: The Expert Recommendation for Implementing Change Taxonomy; JBI: The Joanna Briggs Institute; MMAT: Mixed Method Appraisal Tool.

**Contributorship statement**

All authors were involved in the critically revised manuscript for publication and contributed to the study design and writing. All authors approved the final manuscript.

**Competing interests**

The authors declare that they have no competing interests.

**Funding**

This systematic review will be conducted as a part of the PhD project for the first author, which is funded by Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

**Data sharing statement:**



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Data sharing not applicable as no datasets generated and/or analysed for this study.

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# Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

## Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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| | | Reporting Item | Page Number |
|---|---|---|---|
| **Title** | | | |
| Identification | #1a | Identify the report as a protocol of a systematic review | 1 |
| Update | #1b | If the protocol is for an update of a previous systematic review, identify as such | n/a |

| | | | |
|---|---|---|---|
| **Registration** | | | |
| | #2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | 2 |
| **Authors** | | | |
| Contact | #3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | 1 |
| Contribution | #3b | Describe contributions of protocol authors and identify the guarantor of the review | 12 |
| **Amendments** | | | |
| | #4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | n/a |
| **Support** | | | |
| Sources | #5a | Indicate sources of financial or other support for the review | 12 |
| Sponsor | #5b | Provide name for the review funder and / or sponsor | 12 |
| Role of sponsor or funder | #5c | Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol | 12 |
| **Introduction** | | | |

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|---|---|---|---|
| Rationale | #6 | Describe the rationale for the review in the context of what is already known | 2-3-4-5 |
| Objectives | #7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | 5 |
| **Methods** | | | |
| Eligibility criteria | #8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | 6-7-8 |
| Information sources | #9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | 8 |
| Search strategy | #10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | 8-9 |
| Study records - data management | #11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | 9 |
| Study records - selection process | #11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | 9 |

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|---|---|---|---|
| Study records - data collection process | #11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | 9-10-11 |
| Data items | #12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | 10 |
| Outcomes and prioritization | #13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | 6 &11 |
| Risk of bias in individual studies | #14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | 11 |
| Data synthesis | #15a | Describe criteria under which study data will be quantitatively synthesised | 11 |
| Data synthesis | #15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ) | n/a |
| Data synthesis | #15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | n/a |

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| | | | |
|---|---|---|---|
| Data synthesis | #15d | If quantitative synthesis is not appropriate, describe the type of summary planned | n/a |
| Meta-bias(es) | #16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | n/a |
| Confidence in cumulative evidence | #17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | 11 |

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# Implementation strategies supporting fall prevention interventions in a long-term care facility for older persons: a systematic review protocol

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Manuscripts

# Implementation strategies supporting fall prevention interventions in a long-term care facility for older persons: a systematic review protocol

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## Abstract

**Introduction:** Falls are common among older people in long-term care facilities (LCFs). Falls lead to significant morbidity, mortality, and reduced quality of life among residents. Fall prevention interventions have been shown to reduce falls in LCFs. However, this may not always translate to effectiveness in real-world situations. We will conduct a systematic review (SR) to identify the implementation strategies used in fall prevention interventions in LCF, describing the effectiveness of strategies in terms of key implementation outcomes and falls reduction.

**Methods and analysis**: The search will include scientific papers in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, and published theses. The SR will consider all original research that empirically evaluated or tested implementation strategies to support fall prevention interventions in LCF, published in English or Arabic between 1st January 2001 and 21st August 2021, where data is presented on the implementation strategy (e.g. audit and feedback, champions) and/or implementation outcome (e.g. fidelity). Clinical trials, quasi-



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experimental studies and quality improvement studies will be eligible for inclusion. Two researchers will complete abstract screening, data abstraction and quality assessments, independently. The screening process will be presented using a PRISMA flow diagram. Data will be extracted into a standardised table, including the country, year, authors, type of study, primary clinical outcome (falls rate and/or risk reduction as available), implementation strategy, and implementation outcomes. Implementation strategies will be defined and categorised using the Expert Recommendation for Implementing Change Taxonomy. Implementation outcomes will be defined and categorised using the Implementation Outcomes Taxonomy, and clinical outcomes of the intervention effectiveness for falls preventions will be reported as formulated in each study, with a final narrative synthesis of data.

**Ethics and dissemination:** Ethical approval is not required for this study, and the results will be disseminated via peer-reviewed journals and presented at international conferences.

**Trial registration:** This systematic review was registered on the PROSPERO database; registration number CRD42021239604 (30 April 2021).

**Keywords**: Falls prevention, older person, long-term care, residential care, implementation, feasibility.

**Strength and limitation of this study**:

- This planned review of effective implementation strategies and processes used in falls interventions in LCF will help to bridge the evidence-practice gap and to facilitate successful implementation of these interventions in the real-world clinical practice.
- This planned review will Identify the range of implementation strategies used in fall prevention interventions in LCF, and the process of implementing the intervention.
- Data concerning implementation strategies and outcomes will be categorised and defined using the Expert Recommendation for Implementing Change Taxonomy and the



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implementation outcomes taxonomy frameworks, which are considered seminal in the field of implementation science.

- The planned review will include all the clinical trials types, and it is not applicable to be conducted as a meta-analysis.
- The results of this planned review will be summarised to provide a descriptive review.

**Background and rationale**

Falls are considered a serious health issue, with the World Health Organisation (WHO) reporting that more than 30% of people aged 65 or older experience at least one fall yearly, while 50% of these experience recurrent falls (1). A fall is defined as "an event which results in a person coming to rest inadvertently on the ground or floor or other lower level"(2). Falls can lead to decreased physical activity and loss of independence, increased fear of falling and depression, and reduced confidence and life quality among older people (3,4). The incidence rate of falls in long-term care facilities (LCF) is reported to be 1.7 falls per year per resident, more than three times that of community dwellers (5). Half of all LCF residents experience falls each year, while 25% to 30% suffer physical injuries after falling (6,7). Hip fracture due to falls is estimated to occur in 3% to 5% of residents annually in LCF (8). Fall-related injuries substantially increase morbidity, mortality and hospitalisation rates among residents in LCF (7,9). Indeed, 40% of mortality in residential care is caused by fall injuries (10).

Preventing falls among LCF residents is considered a challenge, and many fall prevention interventions have been described, whether single, multicomponent (standardised) or multifactorial (tailored) interventions (11,12). These latter fall prevention interventions typically include exercise, environmental assessment, staff education and training, medication review, vitamin D supplementation, etc. (3). Several systematic reviews (SR) and meta-analyses (MA) have evaluated the effects and efficacy of fall prevention interventions in LCF (3,4,12–14). An SR and MA of 14 studies conducted up to 2013 reported that the rate of recurrent falls decreased by 21% due to fall prevention interventions, although there was no reduction in the overall number of falls (3). It also



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reported that multifactorial interventions significantly reduced the number of falls and recurrent falls, whilst single and multicomponent interventions did not. This same SR reported that staff education and training as a single intervention had a possibly harmful effect by increasing the fall rate in the intervention groups compared to the control groups (3). A later SR of 36 studies conducted up to 2019 reported that fall prevention interventions have a beneficial effect in reducing falls, the number of residents with one fall during the intervention and follow-up, and the number of recurrent fallers by 27%, 20% and 30%, respectively, in LCF residents (12). It also reported that all multifactorial interventions effectively reduced the number of fallers, while single interventions substantially decreased the number of recurrent fallers. As a single component, exercise is effective, reducing by 36% the number of fallers and by 41% the number of recurrent fallers. Some single interventions involving staff education and training demonstrated benefits in terms of reducing falls and recurrent falls, whereas others were not effective (12).

Existing evidence indicates to a degree how multifactorial interventions effectively prevent falls in LCF (15,16). A multidimensional or multi-component intervention simultaneously targets several dimensions such as environmental modification, medication review, staff education, and physical therapy. The specific term “multifactorial intervention” is used when the intervention is tailored to the recipient’s needs rather than applied in a standardised way to all participants (9). Implementing these interventions and other kinds of falls preventions interventions in a ‘real-world’ clinical situation may be complex and challenging, because they contain many components and require change at multiple levels. ‘Real-world’ clinical practice is different to a clinical trial setting. In the latter, there are strict participant inclusion and exclusion criteria, sometimes dedicated supports to implement the intervention, which is usually of a limited duration, and other interventions are not usually occurring simultaneously. In everyday (or “real-world”) clinical practice, the receiving system/staff usually has not volunteered for the intervention, may have multiple competing priorities and demands, including other interventions being implemented simultaneously, and there may be more heterogeneity in staff, patients and systems. In an SR of mixed-method studies which



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aimed to identify factors that limited fall programme success in LCF, there were 27 barriers, most commonly: staff feeling overwhelmed, helpless, staffing issues, limited knowledge and skills, frustration and concern about the ability to control fall management (15). The SR authors recommended staff training to improve communication, knowledge and skill, which are modifiable factors when developing a fall prevention programme (15), and that multifactorial intervention should be addressed barriers to implementation.

The intervention's effectiveness might be influenced by the implementation strategy used, defined as "the methods and techniques used to enhance the adoption, implementation, and sustainability of a clinical programme or practice"(17), and how well the intervention is implemented. The broader implementation science literature highlights the effectiveness of some implementation strategies (e.g. audit and feedback, educational outreach, practice facilitation, local opinion leaders, etc.)(18). The implementation outcomes are considered precursors to service and health outcomes and defined as "the effects of deliberate and purposive actions to implement new treatments, practices, and services"(19), as they give a good indication of the extent and quality of implementation of the intervention. However, all previous SRs (3,4,12–14) reported only sparsely on the implementation strategies or outcomes of the fall prevention interventions. Therefore, determining the process and success of falls prevention intervention implementation is a key issue to be addressed.

Implementation strategies can be described in different ways in different studies, which makes comparison or synthesis of evidence challenging. The Expert Recommendation for Implementing Change Taxonomy (ERIC) is a comprehensive collection of 73 discrete implementation strategies, with agreed labels and definitions for each of these, organised into nine groups (20,21). The ERIC compilation is based on consensus among experts from clinical practice and implementation science, refining Powell et al.'s original list of implementation strategies from health and mental health literature (20). The ERIC compilation was chosen for this review to facilitate the systematic



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description and reporting of implementation strategies, regardless of differences in terminology between studies.

We aim to address the current evidence-practice gap by identifying the range of implementation strategies used in fall prevention interventions for older people in LCF, and their success as measured by implementation outcomes and falls reduction.

**Research question**

- What implementation strategies have been used in single component or multicomponent/multifactorial fall prevention interventions in LCF to date?
- What implementation strategies are most successful in terms of implementation outcomes and falls prevention?

**Objective**

This SR aims to synthesise the evidence on implementation strategies used to support falls prevention interventions (including RCTs of all types, quality improvement projects and quasi-experimental studies) as follows:

- Identify the range of implementation strategies used in fall prevention interventions in LCF, and the process of implementing the intervention.
- Synthesise the effectiveness of the strategies used to implement fall prevention interventions in LCF.

**Methods**

This review was registered on the International Prospective Register of Systematic Reviews (PROSPERO) database (registration number CRD42021239604). The reporting of the review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol (PRISMA-P). The PICO framework was used for framing and reporting the inclusion and exclusion criteria for this protocol.



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**Types of participants/population**

This review will consider all the studies that had an intervention directed to all staff members who worked at LCF for older people aged 65 years and older and reside in LCF. LCFs are defined as "residential facilities that provide 24-hour-a-day surveillance, personal care, and limited clinical care for persons who are typically elderly and infirm" (12). This includes nursing homes and care homes. We will exclude studies reporting on specific populations in LCF; e.g. those residing in long-stay mental health facilities, and studies involving only people with cognitive issues, intellectual disability, etc. We will include studies aimed to mixed populations in LCF, where the data related to older people can be separated.

**Types of interventions**

This review will consider all studies aimed at preventing falls, whether designed as single, multi-component or multifactorial interventions, and whether delivered by multidisciplinary teams or by a single discipline. We will include studies that describe the use of any implementation strategy, such as champions/local leaders, audit/feedback, education materials, workshops, etc. Strategies will be defined using the Expert Recommendation for Implementing Change Taxonomy (ERIC) (20,21)

**Comparisons**

This will include any other interventions or usual care.

**Types of outcomes**

This review will consider two main outcome types: implementation outcome, which measures the success of the implementation process; and focussed patient-related outcomes, i.e. falls rate and risk (19). There will not be restrictions of the studies based on the included outcomes, as long as the implementation strategy is included, but only these two outcome types will be presented. Our primary interest is the implementation outcomes which include feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration of the



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intervention, as per Proctor et al. (19). We are also interested in the effect of the intervention on falls reduction, noting that this is variably reported in intervention studies as falls risk reduction, falls rate reduction, time to first fall, etc. We will not differentiate between injurious and non-injurious falls, and will not present data on secondary outcomes such as medication use or mortality rates.

**Types of studies**

We will consider all intervention studies involving experimental and quasi-experimental designs, including randomised controlled trials (single site, cluster), feasibility studies for randomised controlled trials, quality improvement empirical studies, and pre-post-test empirical designs. Our primary focus is the implementation strategy and process, and the success of this in terms of implementation outcomes. Therefore, as well as quantitative studies, we anticipate including some mixed methods and qualitative studies that accompany or arise from RCTs or pre-post studies, where these explain the implementation success or describe the implementation strategy in more detail. All relevant studies published in English or Arabic in the last 20 years (beginning 1st January 2001) will be included, noting that implementation science is a relatively new field so that we expect little data to be available prior to 2000.

*Table 1: Inclusion and exclusion criteria according to the PICO framework*

| **PICO framework** | **Inclusion criteria** | **Exclusion criteria** |
|---|---|---|
| **Population** | • All staff members in Long Term Care Facilities (LCFs) working with older people (aged 65 and above).<br>• Older LCF populations; or mixed LCF populations where the older population is separately studied<br>• Mixed settings can be included only if LCF-related data is separately reported. | • Intervention is not directed at the staff of the LCF<br>• Studies included only individuals aged under 65, or where data is not separately reported for the older people within a mixed-age population<br>• Studies based outside of an LCF<br>• Studies relating only to specific populations in LCF, e.g. long-stay mental health residents, people with |



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| | | |
|---|---|---|
| | | cognitive issues, intellectual disability, etc<br>• Non-English, non-Arabic language<br>• Published before 2001 |
| **Interventions** | • Fall prevention interventions, whether single component or multifactorial/multicomponent, where there is an implementation strategy or implementation process described | • Studies where the implementation strategy or process is not described |
| **Comparator** | Usual care or other interventions | There will be no restriction on the comparator used in eligible studies |
| **Outcomes** | • Implementation outcomes (e.g., adoption, fidelity, etc.)<br>• Patient-related outcomes (i.e., falls risk and/or rate) | Studies must include implementation outcomes and/or falls outcomes to be included |

**Information resource and search strategy**

We will conduct searches for all published interventions studies (as specified above) in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, limited to the last 20 years. The search will also include published theses within the last 20 years, identified using the following grey literature databases:

- OPEN GREY (www.opengrey.eu)
- Open Access Theses and Dissertations (OATD) (www. Oatd.org)
- ProQuest (www. Proquest.com)
- British library EThOS (www.ethos.bl.uk).
- EBSCO Open dissertation (www.ebsco.com)
- RIAN (www.rian.ie)
- LENUS (www. lenushealth.com)
- CORA (www. cora.ucc.ie)

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The process of searching will combine free and controlled terms, including text-specific keywords or medical subject headings (MeSH) terms, and it will be integrated with Boolean operators. The search will be filtered to include studies in the English and Arabic languages (spoken by the authors), in an LCF, all intervention study types, and older people aged 65 years and above (Table 1). A forward and backward citation search of the included studies will be performed. We will also conduct a manual search, using the names of the authors of intervention studies, seeking any linked feasibility or implementation papers.

*Table 2: Search term strategy used in electronic databases*

| | |
|---|---|
| **1- population and settings** | "long-term care setting*" OR "nursing home*" OR "residential care setting*" OR "Residential facilit*" OR "institution care" OR "nursing care" OR "nursing facilit*" OR "continuous care" |
| **2- Intervention/ phenomena of interest** | Faller* OR "accidental fall*" OR falling OR falls OR slip* OR "fall prevention" OR "falls prevention"<br>**AND**<br>Prevent* OR reduc* OR minimi* OR decreas* OR interven* |
| **3- outcome** | Feasib* OR sustain* OR adopt* OR accept* OR appropriate* OR fidelity OR implement* OR uptake* OR adher* OR facilitat*OR barrier* OR accessib* OR penetrat* OR mechanism* OR mediat* OR driv* |
| **Combination with Boolean operators** | 1 AND 2 AND 3 |
| **limitation** | English and Arabic languages<br>Published since 2001 |

**Data management and selection process**

All retrieved articles will be imported and managed using the evidence synthesis software Covidence (www.covidence.org), with full references included. Duplicate studies will be removed using the application. Two independent reviewers will screen the retrieved studies based on titles and abstracts, discussing any discrepancies and agreeing with a list of potential studies for full screening. The two reviewers will then examine the full text of these articles independently to ensure that they fulfil the inclusion criteria. Disagreement about eligibility at this stage will be resolved through



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discussion and consultation with an additional reviewer if necessary. The results of the search strategy and selection of studies for inclusion will be reported using a PRISMA flow diagram.

**Data Extraction**

Two independent reviewers will extract the data regarding the eligible studies. The data will then be compared, and any disagreement will be solved by discussing until a consensus is reached, with input from a third (senior) researcher if needed.

The data extracted will include the following details:

- First Author, Publication year, Country
- Study design; study duration (intervention and follow-up periods)
- Participant eligibility criteria and sample size (e.g., patient criteria, staff criteria); participant data
- Fall intervention characteristics (e.g., type of intervention; usual care or control intervention)
- Implementation theory or framework
- Implementation strategy (type, number, target, temporality, dose)
- Implementation outcomes (e.g., fidelity)
- Clinical outcome (i.e., direct fall-related outcomes described in the included studies such as falls risk, falls rate, time to first fall, etc.,)

**Data synthesis procedure**

Data concerning implementation strategies and outcomes will be categorised and defined using Powell et al.'s and Proctors et al.'s frameworks, which are considered seminal in the field of the implementation science (19,22). Originally, the Powell et al. (2009) framework contained a list of 68 implementation strategies based on a review of healthcare and mental healthcare literature (22). A group of researchers and expert clinicians developed this work, adding to the compilation and generating expert consensus on a common set of terms and definitions (20). These implementation strategies were updated, reaching 73 in number, under the title of the Expert Recommendation for



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Implementing Change Taxonomy (ERIC), and these strategies were grouped into nine categories (20,21). The implementation strategy data for this review will be coded and characterised using the labels and definitions for the 73 distinct strategies and then synthesised under these nine subheadings.

Proctor et al. developed an Implementation Outcome Taxonomy that used a narrative review approach, whereby an expert group from the implementation sciences collated the implementation outcome definitions used by investigators, and determined the similarities and differences between them, determining clear concepts for labelling the implementation process (19). This taxonomy distinguishes implementation outcomes from service or system outcomes (e.g. reduced waiting time) and clinical treatment outcomes (e.g. mortality). It provides definitions of eight implementation outcomes: feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration; these support the labelling and assessment of the implementation process. The implementation outcome data of this review will be coded and synthesised using the Proctor Implementation Outcome Taxonomy.

**Quality appraisal**

Two independent reviewers will assess the quality of the studies included. Any disagreement will be discussed and resolved via a third reviewer. We expect that some intervention studies will embed implementation data within the main clinical outcome paper, and others will report this as a separate paper.

All RCTs and quasi-experimental studies will be assessed using the relevant checklists from the Joanna Briggs Institute (JBI ) Critical Appraisal Tool for determining the quality of the studies (23,24). Each criterion is rated as “Yes” if the criteria are clearly reported, “NO” if not, “Unclear” if there is no information, or “Not Applicable” if not relevant to the study (25).



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The Mixed Method Appraisal Tool (MMAT) will be used to assess the quality of mixed data (26). MMAT consists of five categories based on the study designs. It is rated with a "Yes" if the criteria are clearly met, or a "NO" if not; "can't tell" can be selected if there is unclear information on the reporting relating to the criterion. The authors have suggested calculating an overall score by scoring the presence/absence of each criteria as 1/0 and then dividing the sum of "presence" responses by the number of "relevant criteria" and multiplying this by 100 (27).

**Patient and public involvement**

There has been no direct contribution from patients or the public to the methodological design of this protocol, but the need for this systematic review was raised by clinicians working in residential care facilities. The preliminary results will be reviewed by a member of the public who works in a LCF and by a member of a national support and advocacy service for older people and healthcare patients, who will represent residents and their families. Both of these will review and contextualise the results, supporting the writing of the discussion/conclusion section of the review paper.

**Summary of evidence and reporting of the review**

The results will be summarised to provide a descriptive review of effective implementation strategies for interventions aimed at preventing falls in long-term care facilities, both in terms of leading to successful clinical outcome and in terms of implementation success. As intervention effectiveness is not the main focus of this review, the effects of the interventions will be presented in table format and summarised narratively.

The reporting of the review will follow Systematic Reviews and Meta Analyses (PRISMA) guidelines.

**Abbreviations**

LCF: long-term care facilities; WHO: The World Health Organisation; SR: systematic review; RCT: randomised controlled trial; PROSPERO: The International Prospective Register of Systematic Reviews; PRISMA-P: The Preferred Reporting Items for Systematic Reviews and Meta-Analysis



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Protocol; MeSH: Medical subject headings; ERIC: The Expert Recommendation for Implementing Change Taxonomy; JBI: The Joanna Briggs Institute; MMAT: Mixed Method Appraisal Tool.

**Ethics approval and consent to participate**

Not applicable

**Consent for publication**

Not applicable

**Authors' contributions**

ST, NC, RM, SM, NA conceptualised and designed the protocol, including the search strategy for adapting searches across different electronic databases, study selection, inclusion criteria, data extraction planning, etc. NA drafted the initial manuscript. ST, NC, SM, RM revised the manuscript critically for important intellectual content.

**Availability of data and material**

Not applicable

**Competing interests**

The authors declare that they have no competing interests

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# Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

## Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

Upload your completed checklist as an extra file when you submit to a journal.

In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

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| | | Reporting Item | Page Number |
|---|---|---|---|
| **Title** | | | |
| Identification | #1a | Identify the report as a protocol of a systematic review | 1 |
| Update | #1b | If the protocol is for an update of a previous systematic review, identify as such | n/a |

| | | | |
|---|---|---|---|
| **Registration** | | | |
| | #2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | 2 |
| **Authors** | | | |
| Contact | #3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | 1 |
| Contribution | #3b | Describe contributions of protocol authors and identify the guarantor of the review | 12 |
| **Amendments** | | | |
| | #4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | n/a |
| **Support** | | | |
| Sources | #5a | Indicate sources of financial or other support for the review | 12 |
| Sponsor | #5b | Provide name for the review funder and / or sponsor | 12 |
| Role of sponsor or funder | #5c | Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol | 12 |
| **Introduction** | | | |

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|---|---|---|---|
| Rationale | #6 | Describe the rationale for the review in the context of what is already known | 2-3-4-5 |
| Objectives | #7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | 5 |
| **Methods** | | | |
| Eligibility criteria | #8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | 6-7-8 |
| Information sources | #9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | 8 |
| Search strategy | #10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | 8-9 |
| Study records - data management | #11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | 9 |
| Study records - selection process | #11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | 9 |

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|---|---|---|---|
| Study records - data collection process | #11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | 9-10-11 |
| Data items | #12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | 10 |
| Outcomes and prioritization | #13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | 6 &11 |
| Risk of bias in individual studies | #14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | 11 |
| Data synthesis | #15a | Describe criteria under which study data will be quantitatively synthesised | 11 |
| Data synthesis | #15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ) | n/a |
| Data synthesis | #15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | n/a |

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|---|---|---|---|
| Data synthesis | #15d | If quantitative synthesis is not appropriate, describe the type of summary planned | n/a |
| Meta-bias(es) | #16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | n/a |
| Confidence in cumulative evidence | #17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | 11 |

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BMJ Open

# Implementation strategies supporting fall prevention interventions in a long-term care facility for older persons: a systematic review protocol

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Manuscripts

# Implementation strategies supporting fall prevention interventions in a long-term care facility for older persons: a systematic review protocol

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## Abstract

**Introduction:** Falls are common among older people in long-term care facilities (LCFs). Falls lead to significant morbidity, mortality, and reduced quality of life among residents. Fall prevention interventions have been shown to reduce falls in LCFs. However, this may not always translate to effectiveness in real-world situations. We will conduct a systematic review (SR) to identify the implementation strategies used in fall prevention interventions in LCF, describing the effectiveness of strategies in terms of key implementation outcomes and falls reduction.

**Methods and analysis**: The search will include scientific papers in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, and published theses. The SR will consider all original research that empirically evaluated or tested implementation strategies to support fall prevention interventions in LCF, published in English or Arabic between 1st January 2001 and 31st December 2021, where data is presented on the implementation strategy (e.g. audit and feedback, champions) and/or implementation outcome (e.g. fidelity). Clinical trials, quasi-



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experimental studies and quality improvement studies will be eligible for inclusion. Two researchers will complete abstract screening, data abstraction and quality assessments, independently. The screening process will be presented using a PRISMA flow diagram. Data will be extracted into a standardised table, including the country, year, authors, type of study, primary clinical outcome (falls rate and/or risk reduction as available), implementation strategy, and implementation outcomes. Implementation strategies will be defined and categorised using the Expert Recommendation for Implementing Change Taxonomy. Implementation outcomes will be defined and categorised using the Implementation Outcomes Taxonomy, and clinical outcomes of the intervention effectiveness for falls preventions will be reported as formulated in each study, with a final narrative synthesis of data.

**Ethics and dissemination:** Ethical approval is not required for this study, and the results will be disseminated via peer-reviewed journals and presented at international conferences.

**Trial registration:** This systematic review was registered on the PROSPERO database; registration number CRD42021239604 (30 April 2021).

**Keywords**: Falls prevention, older person, long-term care, residential care, implementation, feasibility.

**Strength and limitation of this study**:

- This planned review of effective implementation strategies and processes used in falls interventions in LCF will help to bridge the evidence-practice gap and to facilitate successful implementation of these interventions in the real-world clinical practice.
- This planned review will Identify the range of implementation strategies used in fall prevention interventions in LCF, and the process of implementing the intervention.
- Data concerning implementation strategies and outcomes will be categorised and defined using the Expert Recommendation for Implementing Change Taxonomy and the



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implementation outcomes taxonomy frameworks, which are considered seminal in the field of implementation science.

- The planned review will include all the clinical trials types, and it is not applicable to be conducted as a meta-analysis.
- The results of this planned review will be summarised to provide a descriptive review.

**Background and rationale**

Falls are considered a serious health issue, with the World Health Organisation (WHO) reporting that more than 30% of people aged 65 or older experience at least one fall yearly, while 50% of these experience recurrent falls (1). A fall is defined as "an event which results in a person coming to rest inadvertently on the ground or floor or other lower level"(2). Falls can lead to decreased physical activity and loss of independence, increased fear of falling and depression, and reduced confidence and life quality among older people (3,4). The incidence rate of falls in long-term care facilities (LCF) is reported to be 1.7 falls per year per resident, more than three times that of community dwellers (5). Half of all LCF residents experience falls each year, while 25% to 30% suffer physical injuries after falling (6,7). Hip fracture due to falls is estimated to occur in 3% to 5% of residents annually in LCF (8). Fall-related injuries substantially increase morbidity, mortality and hospitalisation rates among residents in LCF (7,9). Indeed, 40% of mortality in residential care is caused by fall injuries (10).

Preventing falls among LCF residents is considered a challenge, and many fall prevention interventions have been described, whether single, multicomponent (standardised) or multifactorial (tailored) interventions (11,12). These latter fall prevention interventions typically include exercise, environmental assessment, staff education and training, medication review, vitamin D supplementation, etc. (3). Several systematic reviews (SR) and meta-analyses (MA) have evaluated the effects and efficacy of fall prevention interventions in LCF (3,4,12–14). An SR and MA of 14 studies conducted up to 2013 reported that the rate of recurrent falls decreased by 21% due to fall prevention interventions, although there was no reduction in the overall number of falls (3). It also



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reported that multifactorial interventions significantly reduced the number of falls and recurrent falls, whilst single and multicomponent interventions did not. This same SR reported that staff education and training as a single intervention had a possibly harmful effect by increasing the fall rate in the intervention groups compared to the control groups (3). A later SR of 36 studies conducted up to 2019 reported that fall prevention interventions have a beneficial effect in reducing falls, the number of residents with one fall during the intervention and follow-up, and the number of recurrent fallers by 27%, 20% and 30%, respectively, in LCF residents (12). It also reported that all multifactorial interventions effectively reduced the number of fallers, while single interventions substantially decreased the number of recurrent fallers. As a single component, exercise is effective, reducing by 36% the number of fallers and by 41% the number of recurrent fallers. Some single interventions involving staff education and training demonstrated benefits in terms of reducing falls and recurrent falls, whereas others were not effective (12).

Existing evidence indicates to a degree how multifactorial interventions effectively prevent falls in LCF (15,16). A multidimensional or multi-component intervention simultaneously targets several dimensions such as environmental modification, medication review, staff education, and physical therapy. The specific term “multifactorial intervention” is used when the intervention is tailored to the recipient’s needs rather than applied in a standardised way to all participants (9). Implementing these interventions and other kinds of falls preventions interventions in a ‘real-world’ clinical situation may be complex and challenging, because they contain many components and require change at multiple levels. ‘Real-world’ clinical practice is different to a clinical trial setting. In the latter, there are strict participant inclusion and exclusion criteria, sometimes dedicated supports to implement the intervention, which is usually of a limited duration, and other interventions are not usually occurring simultaneously. In everyday (or “real-world”) clinical practice, the receiving system/staff usually has not volunteered for the intervention, may have multiple competing priorities and demands, including other interventions being implemented simultaneously, and there may be more heterogeneity in staff, patients and systems. In an SR of mixed-method studies which



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aimed to identify factors that limited fall programme success in LCF, there were 27 barriers, most commonly: staff feeling overwhelmed, helpless, staffing issues, limited knowledge and skills, frustration and concern about the ability to control fall management (15). The SR authors recommended staff training to improve communication, knowledge and skill, which are modifiable factors when developing a fall prevention programme (15), and that multifactorial intervention should be addressed barriers to implementation.

The intervention's effectiveness might be influenced by the implementation strategy used, defined as “the methods and techniques used to enhance the adoption, implementation, and sustainability of a clinical programme or practice”(17), and how well the intervention is implemented. The broader implementation science literature highlights the effectiveness of some implementation strategies (e.g. audit and feedback, educational outreach, practice facilitation, local opinion leaders, etc.)(18). The implementation outcomes are considered precursors to service and health outcomes and defined as “the effects of deliberate and purposive actions to implement new treatments, practices, and services”(19), as they give a good indication of the extent and quality of implementation of the intervention. However, all previous SRs (3,4,12–14) reported only sparsely on the implementation strategies or outcomes of the fall prevention interventions. Therefore, determining the process and success of falls prevention intervention implementation is a key issue to be addressed.

Implementation strategies can be described in different ways in different studies, which makes comparison or synthesis of evidence challenging. The Expert Recommendation for Implementing Change Taxonomy (ERIC) is a comprehensive collection of 73 discrete implementation strategies, with agreed labels and definitions for each of these, organised into nine groups (20,21). The ERIC compilation is based on consensus among experts from clinical practice and implementation science, refining Powell et al.’s original list of implementation strategies from health and mental health literature (20). The ERIC compilation was chosen for this review to facilitate the systematic



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description and reporting of implementation strategies, regardless of differences in terminology between studies.

We aim to address the current evidence-practice gap by identifying the range of implementation strategies used in fall prevention interventions for older people in LCF, and their success as measured by implementation outcomes and falls reduction.

**Research question**

- What implementation strategies have been used in single component or multicomponent/multifactorial fall prevention interventions in LCF to date?
- What implementation strategies are most successful in terms of implementation outcomes and falls prevention?

**Objective**

This SR aims to synthesise the evidence on implementation strategies used to support falls prevention interventions (including RCTs of all types, quality improvement projects and quasi-experimental studies) as follows:

- Identify the range of implementation strategies used in fall prevention interventions in LCF, and the process of implementing the intervention.
- Synthesise the effectiveness of the strategies used to implement fall prevention interventions in LCF.

**Methods**

This review was registered on the International Prospective Register of Systematic Reviews (PROSPERO) database (registration number CRD42021239604). The reporting of the review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol (PRISMA-P). The PICO framework was used for framing and reporting the inclusion and exclusion criteria for this protocol.



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**Types of participants/population**

This review will consider all the studies that had an intervention directed to all staff members who worked at LCF for older people aged 65 years and older and reside in LCF. LCFs are defined as "residential facilities that provide 24-hour-a-day surveillance, personal care, and limited clinical care for persons who are typically elderly and infirm" (12). This includes nursing homes and care homes. We will exclude studies reporting on specific populations in LCF; e.g. those residing in long-stay mental health facilities, and studies involving only people with cognitive issues, intellectual disability, etc. We will include studies aimed to mixed populations in LCF, where the data related to older people can be separated.

**Types of interventions**

This review will consider all studies aimed at preventing falls, whether designed as single, multi-component or multifactorial interventions, and whether delivered by multidisciplinary teams or by a single discipline. We will include studies that describe the use of any implementation strategy, such as champions/local leaders, audit/feedback, education materials, workshops, etc. Strategies will be defined using the Expert Recommendation for Implementing Change Taxonomy (ERIC) (20,21)

**Comparisons**

This will include any other interventions or usual care.

**Types of outcomes**

This review will consider two main outcome types: implementation outcome, which measures the success of the implementation process; and focussed patient-related outcomes, i.e. falls rate and risk (19). There will not be restrictions of the studies based on the included outcomes, as long as the implementation strategy is included, but only these two outcome types will be presented. Our primary interest is the implementation outcomes which include feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration of the



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intervention, as per Proctor et al. (19). We are also interested in the effect of the intervention on falls reduction, noting that this is variably reported in intervention studies as falls risk reduction, falls rate reduction, time to first fall, etc. We will not differentiate between injurious and non-injurious falls, and will not present data on secondary outcomes such as medication use or mortality rates.

**Types of studies**

We will consider all intervention studies involving experimental and quasi-experimental designs, including randomised controlled trials (single site, cluster), feasibility studies for randomised controlled trials, quality improvement empirical studies, and pre-post-test empirical designs. Our primary focus is the implementation strategy and process, and the success of this in terms of implementation outcomes. Therefore, as well as quantitative studies, we anticipate including some mixed methods and qualitative studies that accompany or arise from RCTs or pre-post studies, where these explain the implementation success or describe the implementation strategy in more detail. All relevant studies published in English or Arabic in the last 20 years (beginning 1st January 2001) will be included, noting that implementation science is a relatively new field so that we expect little data to be available prior to 2000.

*Table 1: Inclusion and exclusion criteria according to the PICO framework*

| **PICO framework** | **Inclusion criteria** | **Exclusion criteria** |
|---|---|---|
| **Population** | • All staff members in Long Term Care Facilities (LCFs) working with older people (aged 65 and above).<br>• Older LCF populations; or mixed LCF populations where the older population is separately studied<br>• Mixed settings can be included only if LCF-related data is separately reported. | • Intervention is not directed at the staff of the LCF<br>• Studies included only individuals aged under 65, or where data is not separately reported for the older people within a mixed-age population<br>• Studies based outside of an LCF<br>• Studies relating only to specific populations in LCF, e.g. long-stay mental health residents, people with cognitive issues, intellectual disability, etc |



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| | | |
|---|---|---|
| | | • Non-English, non-Arabic language<br>• Published before 2001 |
| **Interventions** | • Fall prevention interventions, whether single component or multifactorial/multicomponent, where there is an implementation strategy or implementation process described | • Studies where the implementation strategy or process is not described |
| **Comparator** | Usual care or other interventions | There will be no restriction on the comparator used in eligible studies |
| **Outcomes** | • Implementation outcomes (e.g., adoption, fidelity, etc.)<br>• Patient-related outcomes (i.e., falls risk and/or rate) | Studies must include implementation outcomes and/or falls outcomes to be included |

**Information resource and search strategy**

We will conduct searches for all published interventions studies (as specified above) in electronic databases including PubMed, CINAHL, EMBASE, Psyc INFO, SCOPUS and Web of Science, limited to the last 20 years. The search will also include published theses within the last 20 years, identified using the following grey literature databases:

- OPEN GREY (www.opengrey.eu)
- Open Access Theses and Dissertations (OATD) (www. Oatd.org)
- ProQuest (www. Proquest.com)
- British library EThOS (www.ethos.bl.uk).
- EBSCO Open dissertation (www.ebsco.com)
- RIAN (www.rian.ie)
- LENUS (www. lenushealth.com)
- CORA (www. cora.ucc.ie)

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The process of searching will combine free and controlled terms, including text-specific keywords or medical subject headings (MeSH) terms, and it will be integrated with Boolean operators. The search will be filtered to include studies in the English and Arabic languages (spoken by the authors), in an LCF, all intervention study types, and older people aged 65 years and above (Table 1). A forward and backward citation search of the included studies will be performed. We will also conduct a manual search, using the names of the authors of intervention studies, seeking any linked feasibility or implementation papers.

*Table 2: Search term strategy used in electronic databases*

| | |
|---|---|
| **1- population and settings** | "long-term care setting*" OR "nursing home*" OR "residential care setting*" OR "Residential facilit*" OR "institution care" OR "nursing care" OR "nursing facilit*" OR "continuous care" |
| **2- Intervention/ phenomena of interest** | Faller* OR "accidental fall*" OR falling OR falls OR slip* OR "fall prevention" OR "falls prevention"<br>**AND**<br>Prevent* OR reduc* OR minimi* OR decreas* OR interven* |
| **3- outcome** | Feasib* OR sustain* OR adopt* OR accept* OR appropriate* OR fidelity OR implement* OR uptake* OR adher* OR facilitat*OR barrier* OR accessib* OR penetrat* OR mechanism* OR mediat* OR driv* |
| **Combination with Boolean operators** | 1 AND 2 AND 3 |
| **limitation** | English and Arabic languages<br>Published since 2001 |

**Data management and selection process**

All retrieved articles will be imported and managed using the evidence synthesis software Covidence (www.covidence.org), with full references included. Duplicate studies will be removed using the application. Two independent reviewers will screen the retrieved studies based on titles and abstracts, discussing any discrepancies and agreeing with a list of potential studies for full screening. The two reviewers will then examine the full text of these articles independently to ensure that they fulfil the inclusion criteria. Disagreement about eligibility at this stage will be resolved through



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discussion and consultation with an additional reviewer if necessary. The results of the search strategy and selection of studies for inclusion will be reported using a PRISMA flow diagram.

**Data Extraction**

Two independent reviewers will extract the data regarding the eligible studies. The data will then be compared, and any disagreement will be solved by discussing until a consensus is reached, with input from a third (senior) researcher if needed.

The data extracted will include the following details:

- First Author, Publication year, Country
- Study design; study duration (intervention and follow-up periods)
- Participant eligibility criteria and sample size (e.g., patient criteria, staff criteria); participant data
- Fall intervention characteristics (e.g., type of intervention; usual care or control intervention)
- Implementation theory or framework
- Implementation strategy (type, number, target, temporality, dose)
- Implementation outcomes (e.g., fidelity)
- Clinical outcome (i.e., direct fall-related outcomes described in the included studies such as falls risk, falls rate, time to first fall, etc.,)

**Data synthesis procedure**

Data concerning implementation strategies and outcomes will be categorised and defined using Powell et al.'s and Proctors et al.'s frameworks, which are considered seminal in the field of the implementation science (19,22). Originally, the Powell et al. (2009) framework contained a list of 68 implementation strategies based on a review of healthcare and mental healthcare literature (22). A group of researchers and expert clinicians developed this work, adding to the compilation and generating expert consensus on a common set of terms and definitions (20). These implementation strategies were updated, reaching 73 in number, under the title of the Expert Recommendation for



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Implementing Change Taxonomy (ERIC), and these strategies were grouped into nine categories (20,21). The implementation strategy data for this review will be coded and characterised using the labels and definitions for the 73 distinct strategies and then synthesised under these nine subheadings.

Proctor et al. developed an Implementation Outcome Taxonomy that used a narrative review approach, whereby an expert group from the implementation sciences collated the implementation outcome definitions used by investigators, and determined the similarities and differences between them, determining clear concepts for labelling the implementation process (19). This taxonomy distinguishes implementation outcomes from service or system outcomes (e.g. reduced waiting time) and clinical treatment outcomes (e.g. mortality). It provides definitions of eight implementation outcomes: feasibility, fidelity, adoption, appropriateness, implementation cost, sustainability, acceptability and penetration; these support the labelling and assessment of the implementation process. The implementation outcome data of this review will be coded and synthesised using the Proctor Implementation Outcome Taxonomy.

**Quality appraisal**

Two independent reviewers will assess the quality of the studies included. Any disagreement will be discussed and resolved via a third reviewer. We expect that some intervention studies will embed implementation data within the main clinical outcome paper, and others will report this as a separate paper.

All RCTs and quasi-experimental studies will be assessed using the relevant checklists from the Joanna Briggs Institute (JBI ) Critical Appraisal Tool for determining the quality of the studies (23,24). Each criterion is rated as “Yes” if the criteria are clearly reported, “NO” if not, “Unclear” if there is no information, or “Not Applicable” if not relevant to the study (25).



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The Mixed Method Appraisal Tool (MMAT) will be used to assess the quality of mixed data (26). MMAT consists of five categories based on the study designs. It is rated with a "Yes" if the criteria are clearly met, or a "NO" if not; "can't tell" can be selected if there is unclear information on the reporting relating to the criterion. The authors have suggested calculating an overall score by scoring the presence/absence of each criteria as 1/0 and then dividing the sum of "presence" responses by the number of "relevant criteria" and multiplying this by 100 (27).

**Patient and public involvement**

There has been no direct contribution from patients or the public to the methodological design of this protocol, but the need for this systematic review was raised by clinicians working in residential care facilities. The preliminary results will be reviewed by a member of the public who works in a LCF and by a member of a national support and advocacy service for older people and healthcare patients, who will represent residents and their families. Both of these will review and contextualise the results, supporting the writing of the discussion/conclusion section of the review paper.

**Summary of evidence and reporting of the review**

The results will be summarised to provide a descriptive review of effective implementation strategies for interventions aimed at preventing falls in long-term care facilities, both in terms of leading to successful clinical outcome and in terms of implementation success. As intervention effectiveness is not the main focus of this review, the effects of the interventions will be presented in table format and summarised narratively.

The reporting of the review will follow Systematic Reviews and Meta Analyses (PRISMA) guidelines.

**Abbreviations**

LCF: long-term care facilities; WHO: The World Health Organisation; SR: systematic review; RCT: randomised controlled trial; PROSPERO: The International Prospective Register of Systematic



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Reviews; PRISMA-P: The Preferred Reporting Items for Systematic Reviews and Meta-Analysis Protocol; MeSH: Medical subject headings; ERIC: The Expert Recommendation for Implementing Change Taxonomy; JBI: The Joanna Briggs Institute; MMAT: Mixed Method Appraisal Tool.

**Ethics approval and consent to participate**

Not applicable

**Consent for publication**

Not applicable

**Authors' contributions**

ST, NC, RM, SM, NA conceptualised and designed the protocol, including the search strategy for adapting searches across different electronic databases, study selection, inclusion criteria, data extraction planning, etc. NA drafted the initial manuscript. ST, NC, SM, RM revised the manuscript critically for important intellectual content.

**Availability of data and material**

Not applicable

**Competing interests**

The authors declare that they have no competing interests

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# Reporting checklist for protocol of a systematic review and meta analysis.

Based on the PRISMA-P guidelines.

## Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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| | | Reporting Item | Page Number |
|---|---|---|---|
| **Title** | | | |
| Identification | #1a | Identify the report as a protocol of a systematic review | 1 |
| Update | #1b | If the protocol is for an update of a previous systematic review, identify as such | n/a |

| | | | |
|---|---|---|---|
| **Registration** | | | |
| | #2 | If registered, provide the name of the registry (such as PROSPERO) and registration number | 2 |
| **Authors** | | | |
| Contact | #3a | Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author | 1 |
| Contribution | #3b | Describe contributions of protocol authors and identify the guarantor of the review | 12 |
| **Amendments** | | | |
| | #4 | If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments | n/a |
| **Support** | | | |
| Sources | #5a | Indicate sources of financial or other support for the review | 12 |
| Sponsor | #5b | Provide name for the review funder and / or sponsor | 12 |
| Role of sponsor or funder | #5c | Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol | 12 |
| **Introduction** | | | |

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|---|---|---|---|
| Rationale | #6 | Describe the rationale for the review in the context of what is already known | 2-3-4-5 |
| Objectives | #7 | Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO) | 5 |
| **Methods** | | | |
| Eligibility criteria | #8 | Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review | 6-7-8 |
| Information sources | #9 | Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage | 8 |
| Search strategy | #10 | Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated | 8-9 |
| Study records - data management | #11a | Describe the mechanism(s) that will be used to manage records and data throughout the review | 9 |
| Study records - selection process | #11b | State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis) | 9 |

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|---|---|---|---|
| Study records - data collection process | #11c | Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators | 9-10-11 |
| Data items | #12 | List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications | 10 |
| Outcomes and prioritization | #13 | List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale | 6 &11 |
| Risk of bias in individual studies | #14 | Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis | 11 |
| Data synthesis | #15a | Describe criteria under which study data will be quantitatively synthesised | 11 |
| Data synthesis | #15b | If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ) | n/a |
| Data synthesis | #15c | Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression) | n/a |

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|---|---|---|---|
| Data synthesis | #15d | If quantitative synthesis is not appropriate, describe the type of summary planned | n/a |
| Meta-bias(es) | #16 | Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies) | n/a |
| Confidence in cumulative evidence | #17 | Describe how the strength of the body of evidence will be assessed (such as GRADE) | 11 |

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